AXI4 BFM – Quick Reference

NOTE: As of UVVM v3.x, all shared variables have been made protected. This means that any access to shared variables must be done using get- and set-methods. This documentation has not yet been updated with the methods for accessing these variables, but will be very soon.

BFM

axi_bfm_pkg.vhd

axi_write (awid_value, awaddr_value, awlen_value, awsize_value, awburst_value, awlock_value, awcache_value, awprot_value, awqos_value, awregion_value, awuser_value, wdata_value, wstrb_value, wuser_value, buser_value, bresp_value, msg, clk, axi_if, [scope, [msg_id_panel, [config]]])

Example: axi_write(
```
awid_value    => x"01",
awaddr_value  => x"00000004",
awlen_value   => x"01",
awsize_value  => 4,
awburst_value => INCR,
awlock_value  => NORMAL,
awcache_value => "0000",
awprot_value  => UNPRIVILEGED_UNSECURE_DATA,
awqos_value   => "0000",
awregion_value => "0000",
awuser_value  => x"01",
wdata_value   => t_slv_array'(x"12345678", x"33333333"),
wstrb_value   => t_slv_array'(x"F", x"F"),
wuser_value   => t_slv_array'(x"01", x"01"),
buser_value   => v_buser_value,
bresp_value   => v_bresp_value,
msg           => "Writing data to Peripheral 1",
clk           => clk,
axi_if        => axi_if);
```

Optional parameters (using named association):

- awid_value
- awlen_value
- awsize_value
- awburst_value
- awlock_value
- awcache_value
- awprot_value
- awqos_value
- awregion_value
- awuser_value
- wstrb_value
- wuser_value

Suggested usage: *axi_write(*
```
awaddr_value  => x"00000004",
awlen_value   => x"01",
wdata_value   => t_slv_array'(x"12345678", x"33333333"),
msg           => "Writing data to Peripheral 1");
```
-- Suggested usage requires local overload (see section 5)

axi_read (arid_value, araddr_value, arlen_value, arsize_value, arburst_value, arlock_value, arcache_value, arprot_value, arqos_value, arregion_value, aruser_value, rdata_value, rresp_value, ruser_value, msg, clk, axi_if, [scope, [msg_id_panel, [config, [proc_name]]]])

Example: axi_read(
```
arid_value    => x"01",
araddr_value  => x"00000004",
arlen_value   => x"01",
arsize_value  => 4,
arburst_value => INCR,
arlock_value  => NORMAL,
arcache_value => "0000",
arprot_value  => UNPRIVILEGED_UNSECURE_DATA,
arqos_value   => "0000",
arregion_value => "0000",
aruser_value  => x"01",
rdata_value   => v_rdata_value,
rresp_value   => v_rresp_value,
ruser_value   => v_ruser_value,
msg           => "Read from Peripheral 1",
clk           => clk,
axi_if        => axi_if);
```

Optional parameters (using named association):

- arid_value
- arlen_value
- arsize_value
- arburst_value
- arlock_value
- arcache_value
- arprot_value
- arqos_value
- arregion_value
- aruser_value

Suggested usage: *axi_read(*
```
araddr_value => C_ADDR_IO,
arlen_value  => x"01",
rdata_value  => v_data_out,
msg          => "Read from IO");
```
-- Suggested usage requires local overload (see section 5)

UVVM™

VHDL 2008 only

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axi_check (arid_value, araddr_value, arlen_value, arsize_value, arburst_value, arlock_value, arcache_value, arprot_value, arqos_value, arregion_value, aruser_value, rdata_exp, rresp_exp, ruser_exp, msg, clk, axi_if, [alert_level, [scope, [msg_id_panel, [config]]]])

Example: axi_check(
arid_value => x"01",
araddr_value => x"00000004",
arlen_value => x"01",
arsize_value => 4,
arburst_value => INCR,
arlock_value => NORMAL,
arcache_value => "0000",
arprot_value => UNPRIVILEGED_UNSECURE_DATA,
arqos_value => "0000",
arregion_value => "0000",
aruser_value => x"01",
rdata_exp => t_slv_array'(x"12345678", x"33333333"),
rresp_exp => t_slv_array'("00", "00"),
ruser_exp => t_slv_array'(x"00", x"00"),
msg => "Check data from Peripheral 1",
clk => clk,
axi_if => axi_if);

Optional parameters (using named association):

- arid_value
- arlen_value
- arsize_value
- arburst_value
- arlock_value
- arcache_value
- arprot_value
- arqos_value
- arregion_value
- aruser_value
- rresp_exp
- ruser_exp

Suggested usage: *axi_check(*
araddr_value => C_ADDR_IO,
arlen_value => "01",
rdata_exp => t_slv_array'(x"12345678", x"33333333"),
msg => "Checking data from Peripheral 1");
-- Suggested usage requires local overload (see section 5)

init_axi_if_signals (addr_width, data_width, id_width, user_width)

Example: axi_if <= init_axi_if_signals(addr_width, data_width, id_width, user_width);

BFM Configuration record ´t_axi_bfm_config´

Record element	Type	C_AXI_BFM_CONFIG_DEFAULT	Description
general_severity	t_alert_level	ERROR	Severity level for various checks of AXI transactions.
max_wait_cycles	natural	1000	Used for setting the maximum cycles to wait before an alert is issued when waiting for ready and valid signals from the DUT.
max_wait_cycles_severity	t_alert_level	TB_FAILURE	The above timeout will have this severity
clock_period	time	-1 ns	Period of the clock signal.
clock_period_margin	time	0 ns	Input clock period margin to specified clock_period
clock_margin_severity	t_alert_level	TB_ERROR	The above margin will have the severity
setup_time	time	-1 ns	Setup time for generated signals. Suggested value is clock_period/4. An alert is reported if setup_time exceed clock_period/2.
hold_time	time	-1 ns	Hold time for generated signals. Suggested value is clock_period/4. An alert is reported if hold_time exceed clock_period/2.
bfm_sync	t_bfm_sync	SYNC_ON_CLOCK_ONLY	When set to SYNC_ON_CLOCK_ONLY the BFM will enter on the first falling edge, estimate the clock period, synchronise the output signals and exit ¼ clock period after a succeeding rising edge. When set to SYNC_WITH_SETUP_AND_HOLD the BFM will use the configured setup_time, hold_time and clock_period to synchronise output signals with clock edges.
match_strictness	t_match_strictness	MATCH_EXACT	Matching strictness for std_logic values in check procedures. MATCH_EXACT requires both values to be the same. Note that the expected value

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			can contain the don't care operator '-'. MATCH_STD allows comparisons between 'H' and '1', 'L' and '0' and '-' in both values.
num_aw_pipe_stages	natural	1	Write Address Channel pipeline steps
num_w_pipe_stages	natural	1	Write Data Channel pipeline steps
num_ar_pipe_stages	natural	1	Read Address Channel pipeline steps
num_r_pipe_stages	natural	1	Read Data Channel pipeline steps
num_b_pipe_stages	natural	1	Response Channel pipeline steps
id_for_bfm	t_msg_id	ID_BFM	The message ID used as a general message ID in the AXI BFM
id_for_bfm_wait	t_msg_id	ID_BFM_WAIT	The message ID used for logging waits in the AXI BFM
id_for_bfm_poll	t_msg_id	ID_BFM_POLL	The message ID used for logging polling in the AXI BFM

BFM non-signal parameters

Name	Type	Example(s)	Default value	Description
awid_value	std_logic_vector	x"01"	0	Identification tag for a write transaction
awaddr_value	unsigned	x"125A"	None	The address of the first transfer in a write transaction
awlen_value	unsigned(7 downto 0)	x"01"	0	The number of data transfers in a write transaction
awsize_value	Integer range 1 to 128	4	4	The number of bytes in each data transfer in a write transaction (Must be a power of two)
awburst_value	t_axburst	INCR	INCR	Burst type, indicates how address changes between each transfer in a write transaction
awlock_value	t_axlock	NORMAL	NORMAL	Provides information about the atomic characteristics of a write transaction
awcache_value	std_logic_vector(3 downto 0)	"0000"	(others=>'0')	Indicates how a write transaction is required to progress through a system
awprot_value	t_axprot	UNPRIVILEGED_UNSECURE_DATA	UNPRIVILEGED_UNSECURE_DATA	Protection attributes of a write transaction. Privilege, security level and access type
awqos_value	std_logic_vector(3 downto 0)	"0000"	(others=>'0')	Quality of Service identifier for a write transaction
awregion_value	std_logic_vector(3 downto 0)	"0000"	(others=>'0')	Region indicator for a write transaction
awuser_value	std_logic_vector	x"01"	(others=>'0')	User-defined extension for the write address channel
wdata_value	t_slv_array	t_slv_array'(x"20D3", x"1234")	None	Array of data values to be written to the addressed registers
wstrb_value	t_slv_array	t_slv_array'("1111", "1111")	(others=>'1') for all words	Array of write strobes, indicates which byte lanes hold valid data. (all '1' means all bytes are updated)
wuser_value	t_slv_array	t_slv_array'(x"00", x"01")	(others=>'0') for all words	Array of user-defined extension for the write data channel
buser_value	std_logic_vector	v_buser_value	None	Output variable containing the user-defined extension for the write response channel
bresp_value	t_xresp	v_bresp_value	None	Output variable containing the write response which indicates the status of a write transaction
arid_value	std_logic_vector	x"01"	(others=>'0')	Identification tag for a read transaction
araddr_value	unsigned	x"125A"	None	The address of the first transfer in a read transaction
arlen_value	unsigned(7 downto 0)	x"01"	(others=>'0')	The number of data transfers in a read transaction
arsize_value	Integer range 1 to 128	4	4	The number of bytes in each data transfer in a read transaction (Must be a power of two)
arburst_value	t_axburst	INCR	INCR	Burst type, indicates how address changes between each transfer in a read transaction
arlock_value	t_axlock	NORMAL	NORMAL	Provides information about the atomic characteristics of a read transaction
arcache_value	std_logic_vector(3 downto 0)	"0000"	(others=>'0')	Indicates how a read transaction is required to progress through a system
arprot_value	t_axprot	UNPRIVILEGED_UNSECURE_DATA	UNPRIVILEGED_UNSECURE_DATA	Protection attributes of a read transaction. Privilege, security level and access type
arqos_value	std_logic_vector(3 downto 0)	"0000"	(others=>'0')	Quality of Service identifier for a read transaction
arregion_value	std_logic_vector(3 downto 0)	"0000"	(others=>'0')	Region indicator for a read transaction
aruser_value	std_logic_vector	x"01"	(others=>'0')	User-defined extension for the read address channel

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rdata_value	t_slv_array	v_rdata_value	None	Output variable containing an array of read data
rresp_value	t_xresp_array	v_rresp_value	None	Output variable containing an array of read responses which indicates the status of a read transfer
ruser_value	t_slv_array	v_ruser_value	None	Output variable containing an array of user-defined extensions for the read data channel
rdata_exp	t_slv_array	t_slv_array'(x"ABCD", x"1234")	None	Array of expected read data values. A mismatch results in an alert 'alert_level'
rresp_exp	t_xresp_array	t_xresp_array'(OKAY, OKAY)	OKAY for all words	Array of expected read responses which indicates the status of a read transfer
ruser_exp	t_slv_array	t_slv_array'(x"01", x"01")	(others=>'0') for all words	Array of expected user-defined extensions for the read data channel
alert_level	t_alert_level	ERROR or TB_WARNING	C_AXI_BFM_CONFIG_DEFAULT.general_severity	Set the severity for the alert that may be asserted by the procedure.
msg	string	"Set state active on peripheral 1"	None	A custom message to be appended in the log/alert.
scope	string	"AXI_BFM"	C_SCOPE ("AXI_BFM")	A string describing the scope from which the log/alert originates. In a simple single sequencer typically "AXI_BFM". In a verification component typically "AXI_VVC ".
msg_id_panel	t_msg_id_panel	shared_msg_id_panel	shared_msg_id_panel	Optional msg_id_panel, controlling verbosity within a specified scope. Defaults to a common message ID panel defined in the UVVM-Util adaptations package.
config	t_axi_bfm_config	C_AXI_BFM_CONFIG_DEFAULT	C_AXI_BFM_CONFIG_DEFAULT	Configuration of BFM behaviour and restrictions. See section 2 for details.

BFM signal parameters

Name	Type	Description
clk	std_logic	The clock signal used to read and write data in/out of the AXI4 BFM.
axi_if	t_axi_if	See table "Signal record 'axi_if'"

Note: All signals are active high. See AXI4 documentation for protocol description.

For more information on the AXI4 signals, please see the AXI4 specification.

Signal record 'axi_if'

Record element	Type
write_address_channel	t_axi_write_address_channel
write_data_channel	t_axi_write_data_channel
write_response_channel	t_axi_write_response_channel
read_address_channel	t_axi_read_address_channel
read_data_channel	t_axi_read_data_channel

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Write address channel record **'t_axi_write_address_channel'**

Record element	Type
write_address_channel	**t_axi_write_address_channel**
awid	std_logic_vector
awaddr	std_logic_vector
awlen	std_logic_vector(7 downto 0)
awsize	std_logic_vector(2 downto 0)
awburst	std_logic_vector(1 downto 0)
awlock	std_logic
awcache	std_logic_vector(3 downto 0)
awprot	std_logic_vector(2 downto 0)
awqos	std_logic_vector(3 downto 0)
awregion	std_logic_vector(3 downto 0)
awuser	std_logic_vector
awvalid	std_logic
awready	std_logic

Read address channel record **'t_axi_read_address_channel'**

Record element	Type
read_address_channel	**t_axi_read_address_channel**
arid	std_logic_vector
araddr	std_logic_vector
arlen	std_logic_vector(7 downto 0)
arsize	std_logic_vector(2 downto 0)
arburst	std_logic_vector(1 downto 0)
arlock	std_logic
arcache	std_logic_vector(3 downto 0)
arprot	std_logic_vector(2 downto 0)
arqos	std_logic_vector(3 downto 0)
arregion	std_logic_vector(3 downto 0)
aruser	std_logic_vector
arvalid	std_logic
arready	std_logic

AXI parameter record types

Type name	Allowed value
t_axburst	FIXED
	INCR
	WRAP
t_axlock	NORMAL
	EXCLUSIVE
t_axprot	UNPRIVILEGED_NONSECURE_DATA
	UNPRIVILEGED_NONSECURE_INSTRUCTION
	UNPRIVILEGED_SECURE_DATA

Write data channel record **'t_axi_write_data_channel'**

Record element	Type
write_data_channel	**t_axi_write_data_channel**
wdata	std_logic_vector
wstrn	std_logic_vector
wlast	std_logic
wuser	std_logic_vector
wvalid	std_logic
wready	std_logic

Write response channel record **'t_axi_write_response_channel'**

Record element	Type
write_response_channel	**t_axi_write_response_channel**
bid	std_logic_vector
bresp	std_logic_vector(1 downto 0)
buser	std_logic_vector
bvalid	std_logic
bready	std_logic

Read data channel record **'t_axi_read_data_channel'**

Record element	Type
read_data_channel	**t_axi_read_data_channel**
rid	std_logic_vector
rdata	std_logic_vector
rresp	std_logic_vector(1 downto 0)
rlast	std_logic
ruser	std_logic_vector
rvalid	std_logic
rready	std_logic

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	UNPRIVILEGED_SECURE_INSTRUCTION
	PRIVILEGED_NONSECURE_DATA
	PRIVILEGED_NONSECURE_INSTRUCTION
	PRIVILEGED_SECURE_DATA
	PRIVILEGED_SECURE_INSTRUCTION
t_xresp	OKAY
	EXOKAY
	SLVERR
	DECERR

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1 BFM procedure details and examples

Procedure	Description

axi_write()	**axi_write(awid_value, awaddr_value, awlen_value, awsize_value, awburst_value, awlock_value, awcache_value, awprot_value, awqos_value, awregion_value, awuser_value, wdata_value, wstrb_value, wuser_value, buser_value, bresp_value, msg, clk, axi_if, [scope, [msg_id_panel, [config]]])**
	
	The axi_write() procedure writes the given data to the given address of the DUT, using the AXI4 protocol. For protocol details, see the AXI4 specification.
	- A log message is written if ID_BFM is enabled for the specified message ID panel.
	
	The procedure reports an alert if:
	- wready does not occur within max_wait_cycles clock cycles (alert level: max_wait_cycles_severity, set in the config)
	- awready does not occur within max_wait_cycles clock cycles (alert level: max_wait_cycles_severity, set in the config)
	- bvalid is not set within max_wait_cycles clock cycles (alert level: max_wait_cycles_severity, set in the config)
	
	Examples:
	`axi_write(`
	`awid_value => x”01”,`
	`awaddr_value => x"00000004",`
	`awlen_value => x"01",`
	`awsize_value => 4,`
	`awburst_value => INCR,`
	`awlock_value => NORMAL,`
	`awcache_value => “0000”,`
	`awprot_value => UNPRIVILEGED_UNSECURE_DATA,`
	`awqos_value => “0000”,`
	`awregion_value => “0000”,`
	`awuser_value => x“01”,`
	`wdata_value => t_slv_array'(x"12345678", x"33333333"),`
	`wstrb_value => t_slv_array'(x"F", x"F"),`
	`wuser_value => t_slv_array'(x"01", x"01"),`
	`buser_value => v_buser_value,`
	`bresp_value => v_bresp_value,`
	`msg => "Writing data to Peripheral 1",`
	`clk => clk,`
	`axi_if => axi_if,`
	`scope => C_SCOPE,`
	`msg_id_panel => shared_msg_id_panel,`
	`config => C_AXI_BFM_CONFIG_DEFAULT);`
	
	`axi_write(`
	`awaddr_value => x"00000004",`
	`wdata_value => t_slv_array'(x"12345678", x"33333333"),`
	`buser_value => v_buser_value,`
	`bresp_value => v_bresp_value,`
	`msg => "Writing data to Peripheral 1");`
	
	Suggested usage (requires local overload, see section 5):

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```
axi_write(C_ADDR_DMA, x"AAAA", "Writing data to DMA");
axi_write(C_ADDR_MEMORY, x"FF", v_data_array, "Writing 256 data words to MEMORY");
```

axi_read()

axi_read(arid_value, araddr_value, arlen_value, arsize_value, arburst_value, arlock_value, arcache_value, arprot_value, arqos_value, arregion_value, aruser_value, rdata_value, rresp_value, ruser_value, msg, clk, axi_if, [scope, [msg_id_panel, [config, [proc_name]]]])

The axi_read() procedure reads data from the DUT at the given address, using the AXI4 protocol. For protocol details, see the AXI4 specification. The read data is placed on the output 'rdata_value' when the read has completed.

- The argument "ext_proc_call" is intended to be used internally, when the procedure is called by axi_check().
- A log message is written if ID_BFM is enabled for the specified message ID panel. This will only occur if the argument proc_name is left unchanged.

The procedure reports an alert if:

- The received rid is different from the transmitted arid_value
- arready does not occur within max_wait_cycles clock cycles (alert level: max_wait_cycles_severity, set in the config)
- rvalid is not set within max_wait_cycles clock cycles (alert level: max_wait_cycles_severity, set in the config)

Examples:

```
axi_read(
  arid_value     => x"01",
  araddr_value   => x"00000004",
  arlen_value    => x"01",
  arsize_value   => 4,
  arburst_value  => INCR,
  arlock_value   => NORMAL,
  arcache_value  => "0000",
  arprot_value   => UNPRIVILEGED_UNSECURE_DATA,
  arqos_value    => "0000",
  arregion_value => "0000",
  aruser_value   => x"01",
  rdata_value    => v_rdata_value,
  rresp_value    => v_rresp_value,
  ruser_value    => v_ruser_value,
  msg            => "Read from Peripheral 1",
  clk            => clk,
  axi_if         => axi_if,
  scope          => C_SCOPE,
  msg_id_panel   => shared_msg_id_panel,
  config         => C_AXI_BFM_CONFIG_DEFAULT);

axi_read(
  araddr_value   => x"00000004",
  rdata_value    => v_rdata_value,
  rresp_value    => v_rresp_value,
  ruser_value    => v_ruser_value,
  msg            => "Read from Peripheral 1",
  clk            => clk,
  axi_if         => axi_if);
```

Suggested usage (requires local overload, see section 5):

```
axi_read(C_ADDR_IO, v_data_out, "Reading from IO device");
axi_read(C_ADDR_MEMORY, x"FF", v_data_array_out, "Reading 256 data words from MEMORY");
```

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axi_check()

axi_check(arid_value, araddr_value, arlen_value, arsize_value, arburst_value, arlock_value, arcache_value, arprot_value, arqos_value, arregion_value, aruser_value, rdata_exp, rresp_exp, ruser_exp, msg, clk, axi_if, [alert_level, [scope, [msg_id_panel, [config]]]])

The axi_check() procedure reads data from the DUT at the given address, using the AXI4 protocol. For protocol details, see the AXI4 specification. After reading data from the AXI4 bus, the read data is compared with the expected data, 'rdata_exp'.

- If the check was successful, and the read data matches the expected data, a log message is written with ID_BFM (if this ID has been enabled).
- If the read data did not match the expected data, an alert with severity 'alert_level' will be reported.

The procedure also report alerts for the same conditions as the axi_read() procedure.

Examples:

```
  axi_check(
    arid_value     => x"01",
    araddr_value   => x"00000004",
    arlen_value    => x"01",
    arsize_value   => 4,
    arburst_value  => INCR,
    arlock_value   => NORMAL,
    arcache_value  => "0000",
    arprot_value   => UNPRIVILEGED_UNSECURE_DATA,
    arqos_value    => "0000",
    arregion_value => "0000",
    aruser_value   => x"01",
    rdata_exp      => t_slv_array'(x"12345678", x"33333333"),
    rresp_exp      => t_xresp_array'(OKAY, OKAY),
    ruser_exp      => t_slv_array'(x"00", x"00"),
    msg            => "Check data from Peripheral 1",
    clk            => clk,
    axi_if         => axi_if,
    alert_level    => ERROR,
    scope          => C_SCOPE,
    msg_id_panel   => shared_msg_id_panel,
    config         => C_AXI_BFM_CONFIG_DEFAULT);

  axi_check(
    araddr_value   => x"00000004",
    rdata_exp      => v_rdata_exp,
    msg            => "Check data from Peripheral 1",
    clk            => clk,
    axi_if         => axi_if);
```

Suggested usage (requires local overload, see section 5):

```
  axi_check(C_ADDR_UART_RX, x"3B", "Checking data in UART RX register");
  axi_check(C_ADDR_MEMORY, x"FF", v_rdata_exp_array, "Checking 256 data words from MEMORY");
```

init_axi_if_signals()

init_axi_if_signals(addr_width, data_width, id_width, user_width)

This function initializes the AXI4 interface. All the BFM outputs are set to zeros ('0') and BFM inputs are set to 'Z'.
Note: This function assumes that awid, bid, arid and rid shares a common width (id_width) and that awuser, buser, aruser, ruser also share a common width (user_width)

Example:

```
    axi_if <= init_axi_if_signals(addr_width, data_width, id_width, user_width);
```

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2 BFM Configuration record

Type name: t_axi_bfm_config

Record element	Type	C_AXI_BFM_CONFIG_DEFAULT	Description
general_severity	t_alert_level	ERROR	Severity level for various checks of AXI transactions.
max_wait_cycles	natural	10	Used for setting the maximum cycles to wait before an alert is issued when waiting for ready and valid signals from the DUT.
max_wait_cycles_severity	t_alert_level	TB_FAILURE	The above timeout will have this severity
clock_period	time	-1 ns	Period of the clock signal.
clock_period_margin	time	0 ns	Input clock period margin to specified clock_period
clock_margin_severity	t_alert_level	TB_ERROR	The above margin will have the severity
setup_time	time	-1 ns	Setup time for generated signals. Suggested value is clock_period/4. An alert is reported if setup_time exceed clock_period/2.
hold_time	time	-1 ns	Hold time for generated signals. Suggested value is clock_period/4. An alert is reported if hold_time exceed clock_period/2.
bfm_sync	t_bfm_sync	SYNC_ON_CLOCK_ONLY	When set to SYNC_ON_CLOCK_ONLY the BFM will always wait for the first falling edge of the clock. Then the interface output signals will be set up immediately. After that, the BFM will wait for the rising edge of the clock, sample or set relevant interface signals (if any), and estimate the clock period. The BFM will deactivate potential interface output signals and exit ¼ clock period after a succeeding rising edge. When set to SYNC_WITH_SETUP_AND_HOLD the BFM will always wait for the configured setup time before the first rising edge of the clock. Then the interface output signals will be set up immediately. After that, the BFM will wait for the rising edge of the clock, and sample or set relevant interface signals (if any). Note that the clock period needs to be configured in this mode. The BFM will deactivate potential interface output signals and exit after the configured hold time after a succeeding rising edge.
match_strictness	t_match_strictness	MATCH_EXACT	Matching strictness for std_logic values in check procedures. MATCH_EXACT requires both values to be the same. Note that the expected value can contain the don't care operator '-'. MATCH_STD allows comparisons between 'H' and '1', 'L' and '0' and '-' in both values.
num_aw_pipe_stages	natural	1	Write Address Channel pipeline steps
num_w_pipe_stages	natural	1	Write Data Channel pipeline steps
num_ar_pipe_stages	natural	1	Read Address Channel pipeline steps
num_r_pipe_stages	natural	1	Read Data Channel pipeline steps
num_b_pipe_stages	natural	1	Response Channel pipeline steps

BETA

id_for_bfm	t_msg_id	ID_BFM	The message ID used as a general message ID in the AXI BFM
id_for_bfm_wait	t_msg_id	ID_BFM_WAIT	The message ID used for logging waits in the AXI BFM
id_for_bfm_poll	t_msg_id	ID_BFM_POLL	The message ID used for logging polling in the AXI BFM

3 Additional Documentation

For additional documentation on the AXI4 standard, please see the AXI4 specification “AMBA® AXI™ and ACE™ Protocol Specification”, available from ARM.

4 Compilation

The AXI4 BFM may only be compiled with VHDL 2008. It is dependent on the UVVM Utility Library (UVVM-Util), which is only compatible with VHDL 2008.
See the separate UVVM-Util documentation for more info. After UVVM-Util has been compiled, the axi_bfm_pkg.vhd BFM can be compiled into any desired library.
See the UVVM Essential Mechanisms located in uvvm_vvc_framework/doc for information about compile scripts.

4.1 Simulator compatibility and setup

See README.md for a list of supported simulators. For required simulator setup see UVVM-Util Quick reference.

BETA

5 Local BFM overloads

A good approach for better readability and maintainability is to make simple, local overloads for the BFM procedures in the TB process.
This allows calling the BFM procedures with the key parameters only
e.g.

```
  axi_write(C_ADDR_PERIPHERAL_1, C_TEST_DATA, "Sending data to Peripheral 1");
```

rather than

```
  axi_write(
    awid_value     => x"01",
    awaddr_value   => x"00000004",
    awlen_value    => x"01",
    awsize_value   => 4,
    awburst_value  => INCR,
    awlock_value   => NORMAL,
    awcache_value  => "0000",
    awprot_value   => UNPRIVILEGED_UNSECURE_DATA,
    awqos_value    => "0000",
    awregion_value => "0000",
    awuser_value   => x"01",
    wdata_value    => t_slv_array'(x"12345678", x"33333333"),
    wstrb_value    => t_slv_array'(x"F", x"F"),
    wuser_value    => t_slv_array'(x"01", x"01"),
    buser_value    => v_buser_value,
    bresp_value    => v_bresp_value,
    msg            => "Writing data to Peripheral 1",
    clk            => clk,
    axi_if         => axi_if,
    scope          => C_SCOPE,
    msg_id_panel   => shared_msg_id_panel,
    config         => C_AXI_BFM_CONFIG_DEFAULT);
```

By defining the local overload as e.g.:

```
  procedure axi_write(
    constant addr_value   : in unsigned;
    constant data_value   : in std_logic_vector;
    constant msg          : in string
  ) is
    variable v_buser_value : std_logic_vector(C_USER_WIDTH-1 downto 0);
    variable v_bresp_value : t_xresp;
  begin
    axi_write(
      awid_value     => x"00",                    -- Setting a default value
      awaddr_value   => addr_value,               -- keep as is
      awlen_value    => x"00",                    -- Set to length=1
      awsize_value   => 4,                        -- Setting a default value
      awburst_value  => INCR,                     -- Setting a default value
```

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```
    awlock_value   => NORMAL,                      -- Setting a default value
    awcache_value  => "0000",                      -- Setting a default value
    awprot_value   => UNPRIVILEGED_UNSECURE_DATA, -- Setting a default value
    awqos_value    => "0000",                      -- Setting a default value
    awregion_value => "0000",                      -- Setting a default value
    awuser_value   => x"01",                       -- Setting a default value
    wdata_value    => data_value,                  -- keep as is
    wstrb_value    => x"f"                         -- Setting a default value
    wuser_value    => x"01",                       -- Setting a default value
    buser_value    => v_buser_value,               -- Assigning to a local variable
    bresp_value    => v_bresp_value,               -- Assigning to a local variable
    msg            => msg,                         -- keep as is
    clk            => clk,                         -- Clock signal
    axi_if         => axi_if,                      -- Signal must be visible in local process scope
    scope          => C_SCOPE,                     -- Setting a default value
    msg_id_panel   => shared_msg_id_panel,         -- Use global, shared msg_id_panel
    config         => C_AXI_BFM_CONFIG_LOCAL);     -- Use locally defined configuration or C_AXI_BFM_CONFIG_DEFAULT
end;
```

Using a local overload like this also allows the following – if wanted:

- Have address value as natural – and convert in the overload
- Set up defaults for constants. May be different for two overloads of the same BFM
- Apply dedicated msg_id_panel to allow dedicated verbosity control

IMPORTANT
This is a simplified Bus Functional Model (BFM) for AXI4.
The given BFM complies with the AXI4 protocol and thus allows a normal access towards an AXI4 interface. This BFM is not AXI4 protocol checker.
For a more advanced BFM please contact Bitvis AS at support@bitvis.no

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